LIVING THINGS

Food Chains and Food Webs

Producers and consumers, predators and prey, food webs and what happens when living things are removed



grass → deer → tiger

In this pack

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This pack belongs to: _____ **Class:** _____

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Food chains and food webs

Name: Class: Date:

Learning goals

- Show feeding relationships as food chains, in words and diagrams.
- Use the words producer, consumer, predator and prey.
- Build food webs and predict what happens when a living thing is removed.

What did you eat today? Perhaps rice, dal and fish. The rice grew using light from the Sun. The fish ate tiny plants in the water. So, in a way, your lunch came from the Sun! Every living thing needs **energy** from food, and food chains show how that energy passes from one living thing to the next.



grass *producer* → **spotted deer** *consumer (prey)* → **tiger** *consumer (predator)*

A Sundarbans food chain. The arrow means "is eaten by" and shows the way the energy goes.

Producers

Every food chain begins with a **plant**. Plants are **producers**: they **make their own food** using energy from **sunlight**, plus water and air. Grass, rice, mangrove leaves and the tiny green **algae** in ponds are producers.

Consumers

Animals cannot make their own food. They are **consumers**: they get energy by eating plants or other animals.

- **Herbivores** eat only plants (deer, cows, caterpillars).
- **Carnivores** eat only animals (tigers, kingfishers).
- **Omnivores** eat plants and animals (monkeys, crows, people).

Predators and prey

A **predator** hunts and eats other animals. The animal that is eaten is the **prey**. In the chain above, the tiger is the predator and the deer is its prey. An animal can be both: a frog is a predator of grasshoppers but prey for a snake.

Reading and writing food chains

Always start with the producer and draw the arrows **towards the eater**:

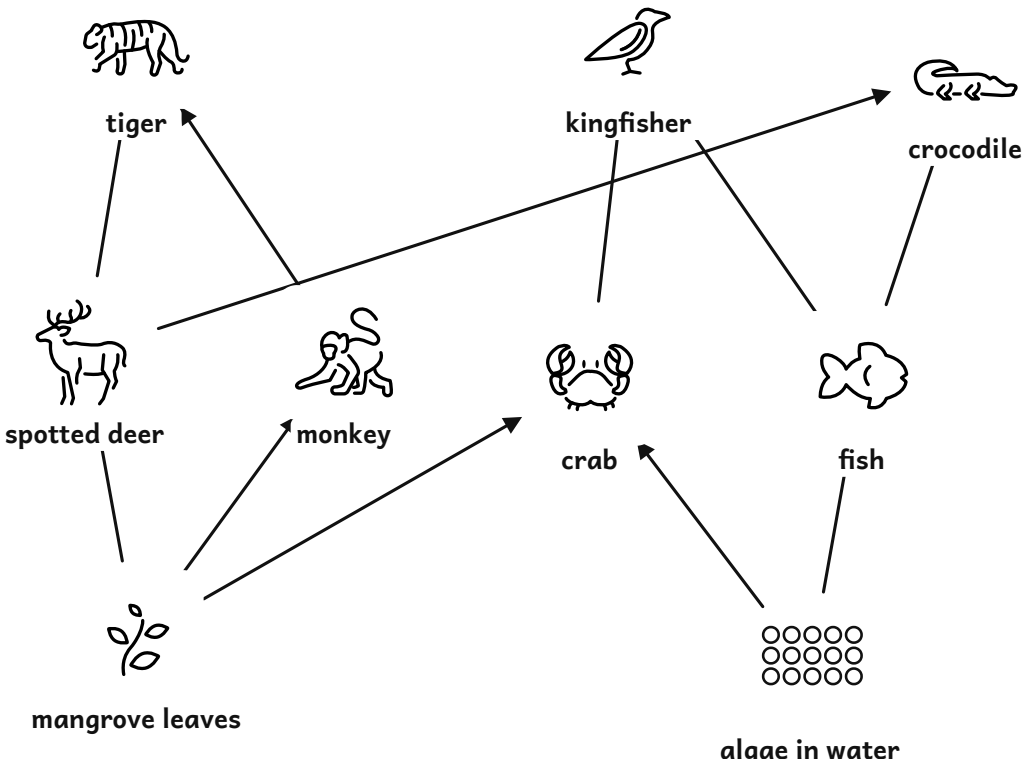
rice plant → grasshopper → frog → snake → eagle

Remember

The arrow does **not** mean "eats". It means "**is eaten by**" or "gives energy to". It points from the food to the animal that eats it.

Food webs

Most animals eat more than one kind of food, and are eaten by more than one predator. When we join many food chains together we get a **food web**. A food web shows all the feeding relationships in a habitat.



A Sundarbans food web.

What if one is removed?

Everything in a food web is connected. If one living thing is removed or dies out, others are affected:

- Its **predators** have less food. They may go hungry, move away or eat more of something else.
- Its **prey** may increase, because fewer of them are being eaten.
- Then the prey may eat too much of **their** food, and so on through the web.

Bangladesh connection

In the Sundarbans, if too many spotted deer are hunted by people, tigers have less food. Hungry tigers may then come closer to villages looking for goats and cows. Protecting the deer also helps protect people and tigers.

Try this at home

Write a food chain for your dinner tonight. Start with the producer. For example: grass → cow → milk → me!

Key words

food chain	shows what eats what, starting with a plant
producer	a plant that makes its own food using sunlight
consumer	an animal that eats other living things
predator	an animal that hunts other animals
prey	an animal that is hunted and eaten
herbivore	eats only plants
carnivore	eats only animals
food web	many food chains joined together

FACT CHECK · A5 · FOOD CHAINS

Who eats whom?

★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Build food chains.
- Use food chain vocabulary.

1 Put the living things in order to make a food chain. Draw the arrows. [3]


snake


rice plant


eagle


grasshopper


frog

2 Use the food chain in question 1. [4]

Question	Answer
Which is the producer?	
Which is a herbivore?	
Name a predator and its prey.	
Where does the energy come from first?	

3 Sort the animals. [9]

SORT THESE

caterpillar tiger deer cow person monkey crow crocodile kingfisher

Herbivore	Carnivore	Omnivore

4 **True or false?** [3]

Statement	True	False	If false, correct it
1. Every food chain starts with a plant.			
2. The arrow in a food chain points to the food.			
3. An animal can be both a predator and prey.			

Exploring a food web

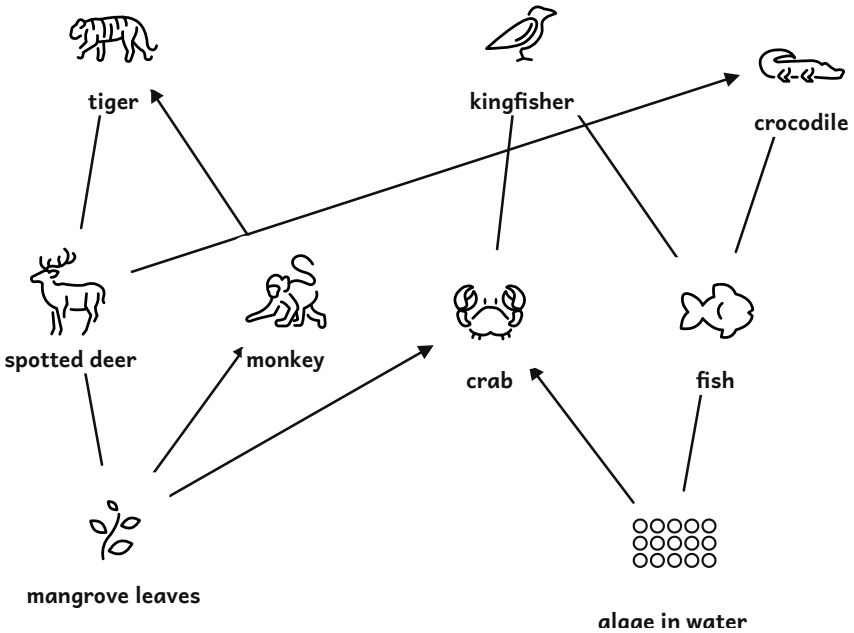
★★☆ Grade 4

Name: Class: Date:

Learning goals

- Read a food web.
- Predict effects of removing a living thing.

1 **Look at the Sundarbans food web.** [7]



a) Write two food chains from the web.

b) Name an animal that eats more than one kind of food.

c) What would happen to the kingfishers if all the fish died? Explain.

d) What would happen to the crabs if all the kingfishers died?

2 **Why do almost all food chains begin with a green plant?** [2]

INVESTIGATE · A5 · FOOD CHAINS

Counting a rice field food chain

★★☆
Grade 4

Name: Class: Date:

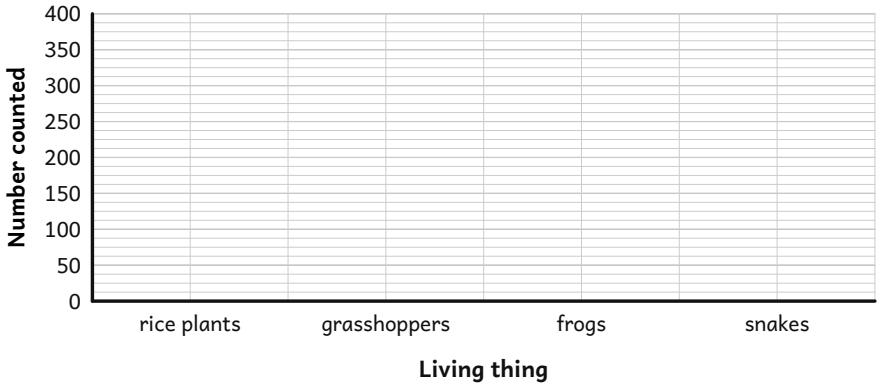
Learning goals

- Record and present survey data.
- Spot a pattern.
- Explain the pattern.

Scientists counted the living things in one small part of a rice field (example data).

1 **Draw a bar chart.** [3]

Living thing	rice plants	grasshoppers	frogs	snakes
Number counted	400	60	8	1



2 **Describe the pattern as you go along the food chain.** [2]

.....

.....

3 **Suggest why there are so few snakes.** [2]

.....

.....

4 **Farmers sprayed a poison that killed most of the frogs. Predict what would happen to:** [3]

a) the grasshoppers

b) the rice plants

c) the snakes

REAL-WORLD CASE STUDY · A5 · FOOD CHAINS

When the vultures disappeared

★★★ Grade 4

Name: Class: Date:

Learning goals

- Use a real food web problem.
- Interpret data about a falling population.
- Evaluate solutions.

Science | Health | Maths | Citizenship

Vultures are large birds that eat the bodies of dead animals. They are nature's cleaners: a group of vultures can clean a dead cow in less than an hour, stopping germs from spreading.

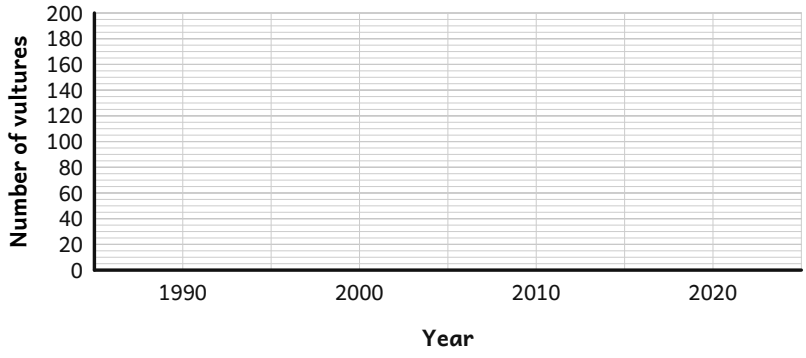
Since the 1990s, the number of vultures in South Asia has fallen by **more than 95 out of every 100**. Scientists found the cause: a medicine called **diclofenac** that was given to sick cattle. When a treated cow died and vultures ate it, the medicine poisoned the vultures.

With fewer vultures, dead animals lay rotting for longer. More stray dogs and rats came to eat them, and diseases can spread more easily. Bangladesh **banned diclofenac** for animals in 2010 and set up **Vulture Safe Zones** in 2014, where vultures are fed safe food and the medicines are checked.

Based on research by vulture conservation groups (e.g. IUCN Bangladesh, SAVE). Numbers below are example data.

1 **Vultures counted at one site (example data).** [3]

Year	1990	2000	2010	2020
Vultures	200	40	10	25



a) Draw the bar chart.

b) What happened after 2010? Suggest why.

.....

.....

2 **Complete the chain of effects.** [3]

- ☐ more stray dogs and rats, and more disease
- ☐ farmers give sick cattle diclofenac
- ☐ dead animals rot for longer
- ☐ vultures eat dead cattle and are poisoned
- ☐ fewer vultures

3 **Who can help vultures? Suggest one action for each.** [3]

Person	Action
a vet	
a farmer	
a learner	

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

V	N	U	W	S	E	P	L	S	U
H	C	C	H	A	I	N	E	R	O
M	O	P	L	A	N	T	S	W	C
U	N	S	E	N	E	R	G	Y	R
P	S	D	V	S	F	W	K	L	R
H	U	V	E	Y	W	G	T	W	S
W	M	P	R	E	D	A	T	O	R
E	E	P	R	O	D	U	C	E	R
B	R	S	U	N	P	R	E	Y	E
D	O	H	V	P	Y	H	C	T	T

chain consumer energy plant predator prey producer sun web

2 In the food chain grass → goat → tiger, which is the prey? [1]

- ◯ grass
- ◯ goat
- ◯ tiger
- ◯ the Sun

3 What do all food chains start with? [1]

- ◯ a predator
- ◯ a herbivore
- ◯ a producer
- ◯ a carnivore

4 Draw a food chain with four living things from a pond. Label the producer, a predator and its prey. [3]

My pond food chain

5 **Challenge. Explain why protecting the producers is the most important way to protect a whole food web.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
build a food chain with arrows the right way	○	○	○
use producer, consumer, predator and prey	○	○	○
sort herbivores, carnivores and omnivores	○	○	○
read a food web	○	○	○
predict what happens when one living thing is removed	○	○	○
explain why vultures matter	○	○	○